

★ PRIMER · UNIT 14

Before the Lecture: Correlation Is Not Enough

Unit 14 · Causality, XAI and Fairness

Watch this first if confounders and conditional probability are hazy

Twenty minutes of groundwork — why two things can move together for a third reason, what a fair share of credit means, and why a regulator will ask you to explain.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before the lecture.
- Confounders and Shapley are here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 14 Sits

The course builds one pipeline; here is this unit's stone

Unit 13

done

Learning by trial and reward.

Unit 14

you are here

Cause, explanation and
fairness.

Unit 15

next

The frontier — attention and
graphs.

Ten units built models that predict. This one asks what they are entitled to claim.

Prediction Against Explanation

A model that predicts well tells you what tends to happen. It does not tell you what would happen if you intervened.

Prediction

- Uses any pattern that works.
- Correlation is enough.
- Cannot survive a change.

Causation

- Asks what if you intervene.
- Needs structure, not just data.
- Survives a policy change.

Every factor that stopped working after publication was a correlation being mistaken for a cause.

The Words the Lecture Assumes

Five terms, and both halves of the unit open up

Word	What it means	Here
Confounder	a third thing driving both	the reason the pattern exists
Spurious	real in the data, not in the world	the factor mirage
Intervention	forcing a variable to a value	what a policy actually does
Counterfactual	what would have happened	the loan you did not grant
Shapley value	a fair share of the credit	how SHAP splits it

A confounder is why ice cream sales predict drownings. Neither causes the other.

Three Rungs of the Causal Ladder

Each rung answers a question the one below cannot

Rung	The question	What it needs
Seeing	what tends to go together?	data alone
Doing	what if I change this?	an experiment, or a model
Imagining	what would have happened?	a full causal model

Machine learning lives on the bottom rung. Most finance questions are asked from the second.

Where Finance Cannot Randomise

You cannot randomly assign a recession to half the world. The clean experiment is simply unavailable.

What medicine does

- Randomise the treatment.
- Compare the two groups.
- Causation follows.

What finance must do

- Reason about structure.
- Use natural experiments.
- Accept weaker claims.

Weaker claims honestly stated beat strong claims that quietly assume a randomised trial.

Dissolving a Fake Signal

A factor predicts returns — until you account for firm size

Model	Factor coefficient	t-statistic	Verdict
Factor alone	0.42	3.10	looks real
Factor + size	0.05	0.31	gone
Size alone	0.51	4.02	this was it

The factor never mattered. Size drove both, and controlling for it dissolves the signal.

A Shapley Value by Hand

Two features, and the fair share of a prediction

The orderings

- A then B: A adds 30.
- B then A: A adds 10.
- Two orders, two answers.

The fair share

- Average across all orders.
- A's Shapley value: 20.
- Order no longer matters.

Averaging over every order is what makes the split fair — and what makes SHAP slow

Why a Good Score Is Not Enough

Three people will ask you a question accuracy cannot answer

The regulator

- Why was this loan refused?
- Under the EU AI Act.
- 'The model said so' will not do.

Your risk officer

- What is it actually using?
- Would it survive a regime change?
- Is it using a proxy for race?

Explanation is not a nicety here. For credit decisions in the EU it is now a legal requirement.

How SHAP Explains One Prediction

Three moves — the same fair-share logic you just did by hand

1 · Start

from the average

What would the model predict
knowing nothing at all?

2 · Add

one feature at a time

How much does the prediction
move when this feature is
revealed?

3 · Average

over all orders

Do that for every ordering, then
take the mean.

The contributions sum exactly to the prediction. That property is why SHAP is trusted.

SHAP Against Unit 7's Importance

They answer different questions, and only one is per-prediction

	Unit 7 importance	SHAP
Scope	the whole model	one single prediction
Sign	no direction	pushed up or down
Consistency	can mislead	guaranteed to be consistent
Cost	free	expensive

For a refused loan you need the second column. The first cannot answer for one person.

Notation You Will Meet

Four symbols carry the whole unit

Symbol	Read it as	It stands for
$\text{do}(x)$	do of x	forcing x, rather than observing it
$P(y \text{do}(x))$	P of y given do x	the effect of intervening
ϕ	phi	a Shapley value — one feature's share
DAG	dag	a diagram of what causes what

The difference between $P(y|x)$ and $P(y|\text{do}(x))$ is the whole of causal inference.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>smf.ols('y ~ factor', data).fit()</code>	the regression that looks convincing
<code>smf.ols('y ~ factor + size', data).fit()</code>	the one that dissolves it
<code>shap.TreeExplainer(model)</code>	fair-share attribution for tree models
<code>explainer.shap_values(X)</code>	one contribution per feature, per row
<code>shap.summary_plot(vals, X)</code>	what the model is really leaning on

Run the second line before believing the first. It is two minutes and saves whole projects.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 15

1

"SHAP shows what causes the outcome"

It shows what the model used. If the model learned a confounder, SHAP will faithfully report it.

2

"We removed race from the features, so the model is fair"

Postcode, income and education carry it anyway. Removing the column removes the evidence, not the bias.

3

"A significant t-statistic means a real effect"

With four hundred factors tried, significance at 5% is what you would expect from noise alone.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1

A factor's t-stat drops from 3.1 to 0.3 after adding size. What was it?

spurious

2

A adds 30 in one order, 10 in the other. Shapley value?

20

3

What does $do(x)$ mean that x alone does not?

intervention

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It separates predicting from explaining

The classic trap, the causal ladder, and a fake signal dissolved on stage.

2

It opens the black box

SHAP as a fair share of credit, computed properly, and set against unit 7's importance.

3

It ends in responsibility

The Apple Card case, the EU AI Act, and what a refused applicant is entitled to be told.

Next: Unit 14 lecture — Cause, Explanation and Fairness